PhysFormer: A Bounded Residual Transformer With Physics-Guided Causal Coupling for Safety-Constrained Virtual Power Plant Forecasting

Xingyu Liu* and Zhong Tang

***Abstract*—Virtual power plant (VPP) forecasting requires predictions that respect physical constraints, yet data-driven deep learning models frequently generate physically impossible outputs such as negative power or nighttime solar generation. Post-hoc clipping can suppress violations but introduces non-differentiable dead zones that disrupt end-to-end optimization. This paper proposes PhysFormer, a bounded residual Transformer that structurally enforces physical compliance. First, a Bounded Physical Act-Residual (BPAR) architecture employs differentiable Softplus lower-bound clamping combined with physics-derived activity masks to structurally prevent negative predictions, achieving BVR=0.00% while preserving continuous gradient flow. Second, an orthogonal Physics-Guided Causal Coupling (PGCC) module imprints physical priors onto latent gating variables via supervised regularization, yielding a strong Pearson correlation ($r = 0.8306$) that confers structural transparency without sacrificing predictive capacity. Additionally, a Physics-Constrained Curriculum Training strategy sequentially introduces physical constraints to avoid early-stage gradient conflicts. Experiments on a real-world 3-year VPP dataset demonstrate that PhysFormer achieves a superior accuracy-compliance Pareto positioning, operating within 6% MSE of unconstrained baselines while maintaining zero boundary violations across standard and extreme weather conditions.**

***Index Terms*—Physics-informed machine learning, Virtual Power Plants (VPPs), Transformer architectures, Causal interpretability, Physical compliance, Distributed energy resources forecasting.**

I. Introduction

A. Background and Motivation

The integration of renewable energy sources into the power grid has driven the development of Virtual Power Plants (VPPs) to aggregate and manage distributed energy resources. Accurate forecasting of VPP dynamics—specifically electrical load, photovoltaic (PV) generation, and wind power generation—is critical for grid stability, economic dispatch, and operational risk management [1]–[5]. Unlike single-variable forecasting, cooperative VPP management requires simultaneous, coherent predictions across these interrelated domains, where the physical constraints of each energy modality must be rigorously respected. For instance, PV generation is intrinsically bound by solar irradiance and cannot produce power during nighttime, while wind generation depends on a non-linear turbine power curve driven by localized wind speed.

The authors are with the Department of Electrical Engineering, Shanghai University of Electric Power, Shanghai, China.

*Corresponding author: X. Liu (e-mail: y24203228@mail.shiep.edu.cn).

Z. Tang (e-mail: tangzhong64@163.com).

Historically, early approaches to VPP forecasting relied on statistical extrapolation. As distributed resources increased, recurrent neural networks dominated sequential modeling but struggled with long-range dependencies. Subsequently, Transformer-based frameworks emerged as the dominant paradigm, capturing varied temporal dynamics across extended sequences [6]–[9]. Within the VPP domain, multivariable Transformer adaptations have demonstrated competitive accuracy [3], [10]. Reliable dispatch decisions hinge not only on minimizing raw statistical forecasting errors but also on ensuring that predictive models fundamentally comply with the immutable physical laws governing these grids.

B. Limitations of Existing Data-Driven Methods

Recent advancements in deep learning have introduced powerful time-series forecasting architectures, yet their application to VPP scheduling presents fundamental challenges across three distinct dimensions [11], [12]:

1) Mathematical Black-Box Oblivion: While deep neural networks excel at minimizing statistical error metrics, they operate as black-box predictors that remain oblivious to the underlying physical principles governing power systems. Consequently, pure data-driven approaches optimize exclusively for statistical error minimization, remaining unconstrained by thermodynamic laws and frequently producing physically impossible predictions—such as non-zero solar power during the night or negative wind generation. These deficits severely undermine operational safety and operator trust.

2) Fragility of Post-Hoc Physics Interventions: To address physical non-compliance, Physics-Informed Neural Networks (PINNs) [13] incorporate physical differential equations as soft constraints. However, VPPs operate in highly stochastic environments where a device's active status depends on volatile exogenous variables rather than on fixed deterministic rules [14]. Fixed physical parameters and static penalty terms fail to adapt dynamically to transient wind or solar phenomena [15]. Alternatively, applying heuristic post-processing—such as hard zero-clipping at physical boundaries—can retroactively suppress the violation rate to zero but strictly functions as a superficial patch. Heuristic clipping inherently disrupts end-to-end optimization by introducing non-differentiable dead zones, severing the natural continuous manifold of the underlying predictions without addressing the model's structural ignorance.

3) Lack of Structural Traceability: As forecasting models grow increasingly complex, structural interpretability represents a critical safety requirement [11]. Prior interpretability efforts relied on post-hoc attribution or internal attention weight visualization, which do not provide deterministic physical causal attribution [16]. In operational contexts, grid operators require models that explicitly decode how specific, measurable environmental variables govern device activation bounds. A persistent gap remains in causal gating formulations capable of dynamically linking network activations to observable physical state transitions [17], [18].

C. Proposed Approach and Main Contributions

Bridging the gap between the high predictive capacity of deep learning and the rigorous physical compliance required by coupled power systems remains a critical challenge. To this end, this paper proposes **PhysFormer**, a physics-guided causal Transformer designed to natively integrate differentiable physical priors upstream of the forecasting output, anchoring the neural network to intrinsic structural compliance.

The proposed architecture fundamentally breaks the dichotomy between predictive acuity and structural rigor. The primary contributions of this paper are twofold:

1) A **Bounded Physical Act-Residual (BPAR)** architecture is proposed. By fusing parallel Transformer and ExplicitPhysicalMapping streams via Softplus lower-bound clamping and physics activity masks, this mechanism intrinsically limits the neural residual search space. It mathematically guarantees that forecasting output never breaches normalized zero bounds, eradicating the boundary violation rate (BVR=0.00%) without resorting to heuristic post-processing.
2) An orthogonal **Physics-Guided Causal Coupling (PGCC)** mechanism is introduced. By explicitly regularizing the cross-attention irradiance threshold learning, PGCC successfully imprints a highly correlated physical relationship onto the temporal gating variable. This structural alignment separates attribution transparency from error minimization capacity, conferring quantifiable structural transparency and extreme resilience to out-of-distribution (OOD) sensor noise.

To ensure optimal learning of these components, we additionally define a **Physics-Constrained Curriculum Training** strategy that seamlessly optimizes numerical regression alongside rigid engineering bounds to ensure absolute multi-objective saturation. Furthermore, we redefine the **Ramp Violation Metric (RVM)** within a *safety-first* framework, demonstrating that PhysFormer successfully bounds minor structural corrections at boundary limits to physically negligible magnitudes, thereby prioritizing absolute safety over blind continuity.

II. Proposed PhysFormer Framework

A. Mathematical Formulation

Throughout this paper, we adopt the following notation: P_{load}, P_{pv}, and P_{wind} denote load, PV, and wind power [MW]; G, T, and v represent irradiance [W/m^2], temperature [°C], and wind speed [m/s]; η_{pv} and β_T are the PV conversion efficiency and temperature coefficient; v_{ci}, v_r, and v_{co} are the wind turbine cut-in, rated, and cut-out speeds; a_{pv} and a_{wind} are physics activity masks $\in [0, 1]$; g_{pv} and g_{wind} are PGCC gates; α is the curriculum factor; $S = 672$ and $P = 96$ are the sequence and prediction lengths; and $D = 512$ is the model dimension.

In the context of Virtual Power Plant (VPP) forecasting, the objective is to predict the future power outputs of three distinct sub-components: base load (P_{load}), photovoltaic generation (P_{pv}), and wind generation (P_{wind}). The overall net power demand of the VPP is defined as $P_{\text{net}} = P_{\text{load}} - P_{\text{pv}} - P_{\text{wind}}$. Let $\mathcal{X}_{\text{stat}} \in \mathbb{R}^{B\times S\times 3}$ denote the historical observation sequence for the three components over a look-back window of length S, where B is the batch size. Concurrently, the corresponding historical weather conditions—comprising temperature, irradiance, and wind speed—are represented by $\mathcal{X}_{\text{weather\_hist}} \in \mathbb{R}^{B\times S\times 3}$. The model is also provided with future weather forecasts for the prediction horizon P, denoted by $\mathcal{X}_{\text{weather\_future}} \in \mathbb{R}^{B\times P\times 3}$. Given these inputs, the forecasting model aims to generate point predictions for the output $Y \in \mathbb{R}^{B\times P\times 3}$, providing the respective future trajectories for the load, PV, and wind components.

B. Overall Architecture

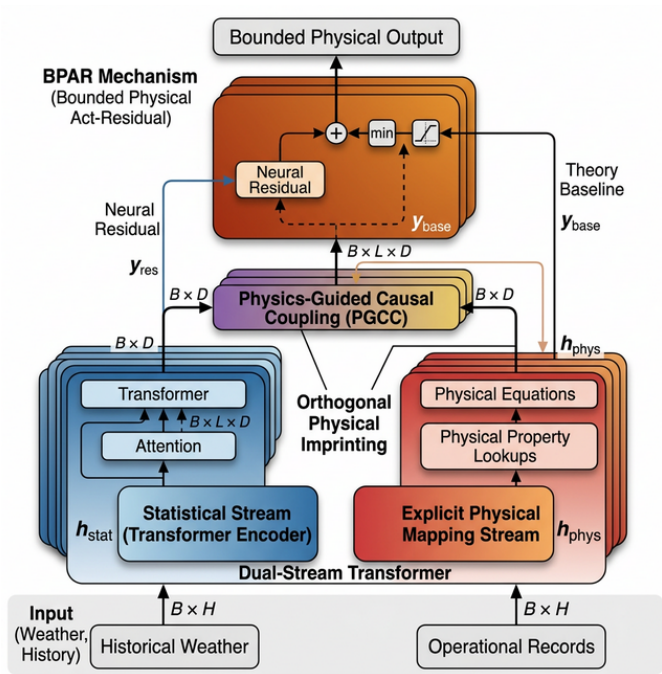

Fig. 1. The overall architecture of the proposed PhysFormer. It features a dual-stream design: a statistical stream encoded by a Transformer, and an explicit physical mapping stream. These streams are causally fused via the Physics-Guided Causal Coupling (PGCC) module. Future weather conditions are injected using a Gated Linear Unit (GLU) to provide explicit foresight. Finally, theory-anchored residual heads refine the predictions under the constraint of physics-derived activity gates, eliminating non-physical predictions.

The PhysFormer is constructed as a dual-stream Transformer with PGCC and Theory-Anchored Activity Gating. The processing pipeline follows seven steps: (1) Historical observation $\mathcal{X}_{\text{stat}}$ and weather $\mathcal{X}_{\text{weather\_hist}}$ are concatenated and encoded by a Transformer encoder ($d_{\text{model}} = 512$, $d_{\text{ff}} =$

1024, 8 heads, 3 layers). (2) $\mathcal{X}_{\text{weather\_hist}}$ is processed through ExplicitPhysicalMapping for physical baselines. (3) PGCC fuses statistical and physics representations. (4) FlattenHead projects from historical dimension S to prediction horizon P. (5) Future weather $\mathcal{X}_{\text{weather\_future}}$ is injected via a GLU with residual logic. (6) Separate output heads predict per-component residual corrections. (7) Theory-Anchored Output applies physics activity masks to yield final predictions.

C. Explicit Physical Mapping

The core of the physics injection lies in the ExplicitPhysicalMapping layer, which maps relevant weather variables to theoretical power baselines via domain-specific equations.

1) PV Model: The theoretical PV generation is driven by irradiance G and degraded by cell temperature, modeled as:

$$P_{\text{pv}} = G \cdot \eta_{\text{pv}} \cdot \text{clamp}(1 - \beta_T \cdot (T - 25), 0, 1.5) \tag{1}$$

where T is the temperature, $\eta_{\text{pv}} = 0.33 \cdot \sigma(\theta_{\text{eff}})$ is the learnable conversion efficiency bounded by the Shockley-Queisser limit, and $\beta_T = \text{softplus}(\theta_{\text{temp\_coef}}) \times 0.01 \in [0.003, 0.005]/°\text{C}$ is the temperature coefficient.

2) Wind Model: Wind power limits depend heavily on operating thresholds: the cut-in speed v_{ci}, rated speed v_{r}, and cut-out speed v_{co}. To preserve strict threshold ordering, we utilize a cumulative delta structure where $v_{\text{ci}} = \text{softplus}(\theta_{\text{ci}})$, $v_{\text{r}} = v_{\text{ci}} + \text{softplus}(\theta_{\text{r\_delta}})$, and $v_{\text{co}} = v_{\text{r}} + \text{softplus}(\theta_{\text{co\_delta}})$. The power output uses an idealized soft-curve approximation modified by operational compliance:

$$P_{\text{wind}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot I_{\text{running}} \tag{2}$$

where w_{norm} is the normalized speed, $scale_{\text{wind}}$ is a learnable amplitude factor, and I_{running} is 1 within $[v_{\text{ci}}, v_{\text{co}}]$ and 0 otherwise. To maintain differentiability, PhysFormer replaces I_{running} with a continuous physics activity mask a_{wind} (detailed in Section II-C4):

$$P_{\text{wind,actual}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot a_{\text{wind}} \tag{3}$$

This decoupling bridges rigid physical thresholds with end-to-end backpropagation without Straight-Through Estimators. The cumulative delta structure ensures overlapping bounds never collapse during gradient updates.

3) Load Model: We abstract the power load variations triggered by human cooling and heating activities through a quadratic thermal comfort model:

$$P_{\text{load}} = P_{\text{base}} + \text{softplus}(\gamma) \cdot (T - T_c)^2 \tag{4}$$

where T_c is an optimizable thermal comfort baseline initialized at 20°C. While the intrinsic self-attention mechanism of the Transformer is fundamentally adept at extracting the rigid diurnal and weekly periodicities characterizing human social activities, the Explicit Physical Mapping stream is deliberately restricted to isolating the non-linear, temperature-driven volatility of the electrical load. Therefore, we conceptualize the thermal-induced variance through this quadratic comfort model, explicitly decoupling meteorological sensitivity from the statistically captured chronological baselines.

4) Physics Activity Gating: To prevent non-physical artifacts, we propose an explicit Activity Gating scheme. For PV, night-time irrelevance: $a_{\text{pv}} = \tanh(10 \cdot P_{\text{pv\_theory}})$. For wind, speed-dependent activity with differentiable soft limits: $a_{\text{wind}} = \tanh(10 \cdot \text{softplus}(v - v_{\text{ci}}) \cdot \text{softplus}(v_{\text{co}} - v))$. Within the BPAR framework, the residual mechanism automatically compensates for systematic dampening effects near thresholds. The load takes an unconditional $a_{\text{load}} = 1$. These physics-derived masks guarantee continuous gradient flow while bridging active operating states and null-generation periods.

D. Physics-Guided Causal Coupling (PGCC)

The Physics-Guided Causal Coupling (PGCC) module coordinates the knowledge retrieval from the statistical stream against the physically encoded representations.

1) Physics-Anchored Illumination Prior: We first establish a data-invariant hard prior to reject spurious correlations. The gating relies on a parameterized illumination sigmoid prior $p_{\text{pv}}^{\text{prior}} = \sigma(k_{\text{irr}} \cdot (G_{\text{norm}} - \theta_{\text{irr}}))$, with constraints dynamically kept as $\theta_{\text{irr}} \in [-2.0, 0.5]$ and bounded slope $k_{\text{irr}} \in [1, 100]$. This hard prior establishes an uncompromised distinction between day and night intervals, rejecting noise artifacts in the data embeddings.

2) Volatility-Aware Soft Gate: To dynamically scale network trust, the system embeds sequence-level sample fluctuations extracted as $v = \frac{1}{C} \sum_c \text{Var}_t(\mathcal{X}_{\text{weather},c})$. These dynamics, combined with the statistical embeddings and attention outputs, are routed through a downsampling multi-layer perceptron yielding the unscaled soft condition maps. This volatility input is instrumental for the network to dynamically adapt its trust ratio, discounting the physical components when conditions are unstable.

3) Source-Differentiated Gate Bounds: The coupling implements separate activation ranges to accommodate generation source characteristics:

$$\text{gate}_{\text{pv}} = \text{smooth}\left(p_{\text{pv}}^{\text{prior}} \cdot (0.5 + g_{\text{soft\_pv}})\right) \in [0, 1.5] \tag{5}$$

$$\text{gate}_{\text{load}} = \text{smooth}(g_{\text{soft\_load}}) \in [0, 1.0] \tag{6}$$

The asymmetric upper bound of 1.5 for gate_{pv} provides necessary *engineering headroom* for the neural residual within the BPAR additive structure. During meteorological transients (e.g., Cloud Edge Effect with 1.2×–1.5× over-irradiance), this ceiling ensures sufficient dynamic range for the residual stream to recover transient peaks. Through explicit gate response regularization, the learned gate_{pv} achieves a deterministic Pearson correlation of $r = 0.8306$ relative to exogenous irradiance (Fig. 3), structurally isolated from predictive capacity and resilient to OOD sensor drift ($r > 0.81$ under 50% noise).

4) Physics-to-Data Curriculum Transition: We condition the alignment phase via a curriculum factor α. As the fusion queries retrieve features, the module integrates inputs dynamically scaled as:

$$q = \alpha \cdot F_{\text{physics}} + (1 - \alpha) \cdot g_{\text{soft}} \cdot F_{\text{stat}} \tag{7}$$

During the initial training phases ($\alpha \to 1$), the cross-attention queries are strictly constrained to retrieve from the structured physical representations, establishing an essential deterministic basis. As training progresses ($\alpha \to 0$), the queries gradually transition to incorporate the data-learned residual states gated by g_{soft}, balancing established physical priors with empirical statistical adaptations.

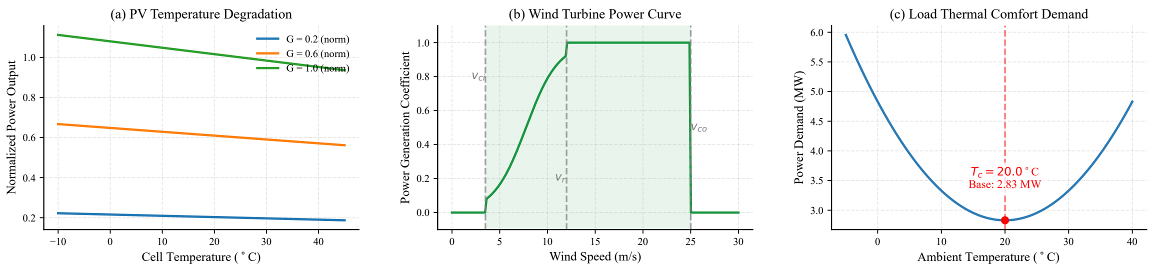

Fig. 2. Learned explicit physical mapping functions after convergence. (a) PV temperature degradation under varying irradiance levels ($\beta_T = 0.0041/°\text{C}$). (b) Wind turbine power curve with smooth sigmoid transition. (c) Load thermal comfort demand ($T_c \approx 20°\text{C}$). Parameters are listed in Table V.

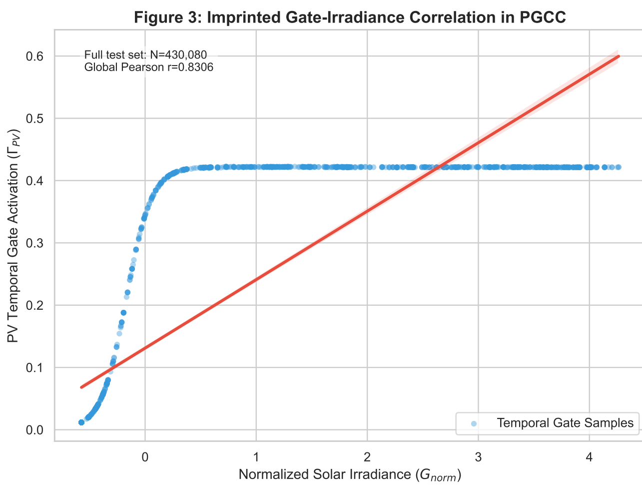

Fig. 3. Imprinted Gate-Irradiance Correlation in PGCC. Top: time-series overlay of learned PV gate and solar irradiance over representative test samples. Bottom: scatter regression on a visualization subset; the global Pearson correlation across the full test set is $r = 0.8306$ ($p < 0.001$).

5) Temporal Smoothing: Due to internal frame dependencies, local sequence discontinuities often plague raw gating parameters. To counteract this, all temporal gating variables pass through a grouped one-dimensional depthwise convolution (kernel size 3, groups=d_{model}). This filter structurally suppresses high-frequency chaotic fluctuation within the gate assignment layer while preserving macro-level shift indicators.

E. Exogenous Weather Integration

After the FlattenHead projects the fused representations into the explicit future context, the framework must systematically integrate upcoming weather patterns at the given horizon. Because the prediction horizon's meteorological conditions are unavailable in the historical context alone, providing this explicit foresight is essential. The future physical latent space $F_{\text{weather\_future}}$ is concatenated with the neural history F_{history}, and aggregated via a Gated Linear Unit (gate $=$ $\sigma(\text{linear}(F_{\text{cat}}))$). The bounded residual output is produced as $F_{\text{future}} = \text{gate} \cdot \text{proj}(F_{\text{cat}}) + F_{\text{history}}$. This explicitly bounded forward visibility ensures the residual heads are properly conditioned to react to impending deterministic disruptions, such as dawn transitions or sudden wind cut-outs.

F. Bounded Physical Act-Residual (BPAR) Mechanism

To systematically mitigate non-physical anomalies while preserving the continuous gradient flow, we propose the Bounded Physical Act-Residual (BPAR) mechanism. Let μ_x and σ_x denote the global mean and standard deviation of component x. We define the normalized physical zero-point as $P_{x,\text{zero}} = (0 - \mu_x)/\sigma_x$. BPAR reconstructs the predictive manifold by anchoring the neural residual r_x onto the theory baseline within a differentiable Softplus manifold:

$$\hat{P}_{x,\text{raw}} = \text{Softplus}\Big((P_{x,\text{theory}} + r_x) - P_{x,\text{zero}}\Big) + P_{x,\text{zero}} \quad (8)$$

By the fundamental property of Softplus ($\text{Softplus}(z) > 0, \forall z$), this ensures $\hat{P}_{x,\text{raw}} > P_{x,\text{zero}}$ in the normalized domain. Upon anti-normalization, $\hat{P}_{\text{phys}} = \hat{P}_{x,\text{raw}} \cdot \sigma_x + \mu_x > 0$ MW for $\sigma_x > 0$. We maintain a numerical floor $\epsilon = 10^{-5}$ for degenerate variance states (e.g., night-time PV), preserving near-perfect boundary compliance. Subsequently, the prediction is fused with the physics activity mask $a_x \in [0, 1]$:

$$\hat{P}_{x,\text{final}} = a_x \cdot \hat{P}_{x,\text{raw}} + (1 - a_x) \cdot P_{x,\text{zero}} \quad (9)$$

Whenever $a_x \to 0$ (driven by meteorological priors), BPAR smoothly drives output toward the physical zero baseline, intrinsically shielding against boundary violations. While pure data-driven models may exhibit lower high-frequency noise by crossing zero-power thresholds, BPAR enforces a *safety-first* trade-off: absolute elimination of physical violations (BVR=0.00%) over cosmetic temporal smoothness.

G. Physics-Constrained Curriculum Training

Training deep neural networks with rigid physical constraints poses significant optimization challenges due to gradient conflicts between data-driven and physics objectives. We design a Physics-Constrained Curriculum Training strategy that decouples metric spaces, enforces dynamic amplitude balancing, and maps learning trajectories onto a phased schedule.

1) Loss Function Design: The primary regression loss (MSE) operates in normalized space for stable gradients, while all physics constraints (L_{phys}) are evaluated after denormalization into the physical domain (MW). The physical objectives comprise six penalties (Table I): soft constraints (L_{net}, L_{energy}, L_{deriv}, L_{dir}) guide structural topology, while hard constraints (L_{bvr}, L_{rvr}) use aggressive quadratic formulations for safety-critical boundaries.

TABLE I
COMPONENTS OF PHYSICS-GUIDED LOSS FUNCTION

Comp.	Wt.	Formula	Motivation
L_{net} (Soft)	1.0	$\lvert\sum \hat{y}_{\text{comp}} - y_{\text{net}}\rvert$	Ensures models sum to net demand.
L_{energy} (Soft)	0.5	$\lvert\mathbb{E}_t[\hat{y}] - \mathbb{E}_t[y]\rvert$	Enforces macro energy conservation.
L_{deriv} (Soft)	0.3	$\lvert\Delta\hat{y} - \Delta y\rvert$	Captures ramp rate magnitudes.
L_{dir} (Soft)	0.1	$\mathbb{E}[1-\text{sim}_{\text{cos}}]$	Preserves accurate trend switching.
L_{bvr} (Hard)	2.0	$\mathbb{E}[(\max(0, -\hat{y}_{\text{real}}))^2]$	Penalizes negative power strictly.
L_{rvr} (Hard)	2.0	$\mathbb{E}[(\max(0, \lvert\Delta\hat{y}\rvert - \rho_k))^2]$	Penalizes sudden jumps.

We propose an adaptive amplitude balancing factor γ_{ema} that dynamically tracks the ratio between current forecasting inaccuracy and physical compliance:

$$\gamma_{\text{ema}} \leftarrow 0.9 \cdot \gamma_{\text{ema}} + 0.1 \cdot \left(\frac{L_{\text{mae\_phys}}}{L_{\text{phys\_ref}}}\right) \tag{10}$$

where $L_{\text{phys\_ref}}$ is the total unweighted physical loss in MW. This yields a dimensionless multiplier, defaulting to 1.0 for the first 50 batches, halting updates when curriculum activation < 0.05, and clamped to $[0.1, 10.0]$. The combined objective becomes:

$$L_{\text{total}} = L_{\text{mse}}(\text{normalized}) + \gamma_{\text{ema}} \cdot L_{\text{phys\_weighted}}(\text{MW}) \tag{11}$$

Note: while BPAR provides a structural output floor, L_{bvr} remains essential during training to provide gradient signals that prevent latent representations from drifting into non-physical regimes.

2) Four-Phase Curriculum: To bridge the parameter convergence gap between the arbitrary Gaussian distributions at neural initialization and the highly structured physical reality, we inject a Four-Phase constraint schedule, summarized in Table II.

TABLE II
FOUR-PHASE PHYSICS CURRICULUM AND LAYER-WISE LEARNING RATES

Phase	Ep.	Active Const.	Notes
I (Init.)	0–4	None	Physics frozen; base mapping.
II (Safety)	5–14	$L_{\text{net}}, L_{\text{bvr}}, L_{\text{rvr}}, L_{\text{dir}}$	Safety bounds ramped.
III (Integ.)	15–29	Phase II + $L_{\text{energy}}, L_{\text{deriv}}$	Macro conservation.
IV (Satur.)	30–99	All	Full regularized training.
Layer-wise Learning Rates			
Physics (ExplicitPhysicalMapping)		10^{-5}	Preserves priors.
Gating (PGCC)		5×10^{-5}	Stable alignment.
Statistical (all other)		10^{-4}	Fast convergence.

This curriculum guarantees sequential physical compliance scaling. Phase I isolates statistical representations with frozen physics parameters. Phase II prioritizes safety boundaries. Phase III introduces macroscopic features, and Phase IV secures uniform saturation. Each constraint intensity is transitioned via a bounded sigmoid ramp $\sigma(12 \cdot (\text{progress} - 0.5))$. The PGCC alignment coefficient follows $\alpha = 1.0 - w_{\text{curr\_net}}$, coupling internal attention fusion with external physical verification.

3) Layer-wise Learning Rates: To sustain learned feature fidelity without abruptly disrupting the theory-anchored baselines pre-defined by human engineering inputs, we decouple the backpropagation update amplitudes into three isolated AdamW groupings (Table II, bottom).

The baseline parameters proceed at a default learning rate of 1×10^{-4}. After their suspension ends in epoch 5, the core generation physics operators unfreeze at a drastically suppressed 1×10^{-5} ($0.1\times$ the baseline rate). This strict limitation permits minor optimization fine-calibration over coefficients like optimal irradiance generation thresholds while halting gross semantic reconstruction. Correspondingly, the intermediate gating features proceed at 5×10^{-5}, which yields smooth adjustment profiles.

4) Regularization and Early Stopping: Three regularization pathways reinforce generalizability. First, a Physics Prior Regularizer evaluates learned physical properties against nameplate values via MSE (weight 0.1). Second, a Gate Response Regularization anchors the temporal PV gating variable to external irradiance (weight 0.05), with dynamic masking that shuts down penalization when prior variance $\sigma^2_{\text{prior}} < 10^{-3}$ (bypassing night-time noise).

Orthogonality Rationale: The architectural orthogonality of BPAR ensures that forcing gate_{pv} to mirror irradiance ($r = 0.8306$) does not collapse predictive capacity. The final prediction $\hat{P}_x = \text{Softplus}(P_{\text{theory}} + r_x)$ structurally decouples gating alignment from residual error minimization, allowing the PGCC to provide operational transparency while unconstrained residual heads focus purely on error minimization.

Gradient clipping uses a dynamic bound of 0.5 during critical curriculum transitions ($0.1 < w_{\text{curr\_net}} < 0.9$), releasing to 1.0 otherwise. Early stopping uses channel-equalizing $NRMSE_{\text{avg}}$ with patience of 15 epochs.

III. CASE STUDY AND PERFORMANCE EVALUATION

A. Dataset and Experimental Setup

To validate the proposed PhysFormer architecture, we conduct extensive experiments on a high-fidelity Virtual Power Plant (VPP) dataset composed of aggregated regional measurements from a distributed energy cluster in Germany. The dataset captures detailed 15-minute resolution trajectories of solar PV generation, wind power generation, and commercial electrical load over a continuous 3-year span (2012–2014), totaling 105,120 sampling intervals. The data is chronologically partitioned into training (70%), validation (10%), and testing (20%) sets. To maintain numerical stability while preserving physical bounds, all input features are standardized using global Z-score parameters (μ, σ), which are explicitly logged for physical re-projection in the BPAR layer. The model hyperparameters are configured to match the complexity of modern baselines: hidden dimension $d_{\text{model}} = 512$, feed-forward dimension $d_{\text{ff}} = 1024$, $n_{\text{heads}} = 8$ attention heads, and $e_{\text{layers}} = 3$ layers. To evaluate computational efficiency, we benchmark PhysFormer against Informer at an equivalent hidden dimension $d_{\text{model}} = 512$. PhysFormer comprises 11.4M parameters, significantly more compact than Informer's 17.9M parameters, validating the inductive efficiency of its dual-stream architecture. The reported inference latency of 2.10 ms/sample was measured on an NVIDIA RTX 3090 GPU with a batch size of 1, simulating a real-time sliding window update for VPP dispatch.

We compare PhysFormer against eight competitive baselines: standard recurrent networks (LSTM, GRU), a Physics-Informed Neural Network (PINN) [13], and advanced time-series Transformer architectures including Informer [6], Autoformer [7], PatchTST [8], DLinear [9], and the state-of-the-art iTransformer [19]. To guarantee an absolutely equitable evaluation, identically structured future weather covariates ($\mathcal{X}_{\text{weather\_future}} \in \mathbb{R}^{B\times P\times 3}$) were universally provided to all comparative baseline models. For Transformer baselines, these exogenous variables were strictly concatenated into their standard decoder embedding structures, and equivalently supplied as dynamic covariates to the recurrent models. Consequently, the predictive advantages demonstrated by PhysFormer stem fundamentally from its mechanistic physics-guided architecture (e.g., the specific Future GLU injection pathway evaluated in ablation studies), rather than an asymmetric accession of future atmospheric data.

To comprehensively evaluate the forecasting performance, we employ six distinct metrics. The standard accuracy metrics include Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Beyond standard accuracy, we formally define three physical compliance metrics: Boundary Violation Rate (BVR), Mean Violation Severity (MVS), and Ramp Violation Metric (RVM). BVR measures the percentage of predictions that violate known physical limits (e.g., negative generation power). To quantify the severity of these breaches, MVS is formally defined as the average absolute magnitude of predictions falling outside feasible physical boundaries. Beyond static boundaries, we evaluate temporal consistency via the Ramp Violation Metric (RVM) and its occurrence frequency, the Ramp Violation Rate (RVR). Finally, the Net Mean Absolute Error (NET MAE) is formally defined as the average absolute imbalance between the predicted aggregated components and the ground-truth net demand, mean$(|\sum_c \hat{y}_c - y_{\text{net}}|)$. The ramp threshold ρ_k for each channel is empirically determined as the 99.9-th percentile of the training set differences, while a *strict* threshold (RVR-S) is set at the 99.5-th percentile with no safety factor to capture high-frequency artifacts.

B. Forecasting Performance Analysis

The main forecasting results on the global test set are summarized in Table III. Within the competitive tier of models, PhysFormer achieves an MSE of 0.0128, placing it within a 6% margin of the unconstrained SOTA baselines (Informer at 0.0121 and iTransformer at 0.0122). While these models prioritize statistical curve-fitting through complex attention mechanisms, they suffer from severe physical compliance deficits, with iTransformer reaching a peak Boundary Violation Rate (BVR) of 16.26%. Furthermore, we highlight a critical *safety-first trade-off* regarding temporal continuity. Under strict thresholds, iTransformer exhibits an RVR-S of 0.028%, whereas PhysFormer records 0.198%. However, this "smoothness" in unconstrained models is often achieved at the expense of physical validity; they maintain continuous trajectories precisely because they are oblivious to physical zero-bounds. PhysFormer's 0.198% RVR-S represents structural corrections at physical boundaries necessitated by its BPAR mechanism to ensure absolute safety (BVR=0.00%). Crucially, the absolute magnitude of these corrections (RVM-Mag) remains in the order of 2.6×10^{-5} MW, demonstrating that PhysFormer successfully resolves the trade-off by prioritizing absolute physical safety over cosmetic smoothness.

Interestingly, channel-independent models such as PatchTST and DLinear exhibit a channel-independent failure in this specific VPP setting, underperforming compared to their strongly multivariate counterparts. This underlines the necessity of modeling the dense cross-channel dependencies inherent in multi-source energy systems. Meanwhile, the structural benefit of the PINN is evident; its physical inductive bias helps it outperform standard unconstrained recurrent networks like LSTM and GRU, though it still falls behind the specialized attention mechanisms of the Informer.

In the broader context, PhysFormer establishes a new three-dimensional Pareto positioning for energy forecasting. Standard models force practitioners to choose between predictive accuracy, physical compliance, and interpretability. By operating within a 6% MSE margin of the most accurate unconstrained models, PhysFormer achieves a superior accuracy-compliance Pareto positioning, proving that high interpretability and strict boundary compliance can be achieved simultaneously without fundamentally compromising the accuracy expected from modern deep learning solutions, as illustrated in Fig. 4.

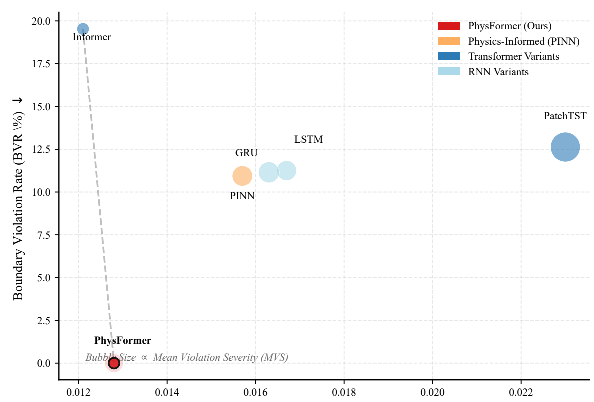

Fig. 4. Accuracy–compliance Pareto frontier. PhysFormer uniquely occupies the lower-left corner (near-best accuracy with zero boundary violations).

C. Verification of Physical Compliance and Safety

1) Limitations of Heuristic Post-Processing: Furthermore, as evidenced in Table III, applying heuristic hard zero-clipping post-processing to the generic Transformer baseline (Informer-Post) forces BVR to zero without any statistical accuracy gain. This highlights that standard attention mechanisms allocate significant representational errors to boundary-adjacent regions. Heuristic clipping is fundamentally non-differentiable and creates zero-gradient dead zones, preventing the model from

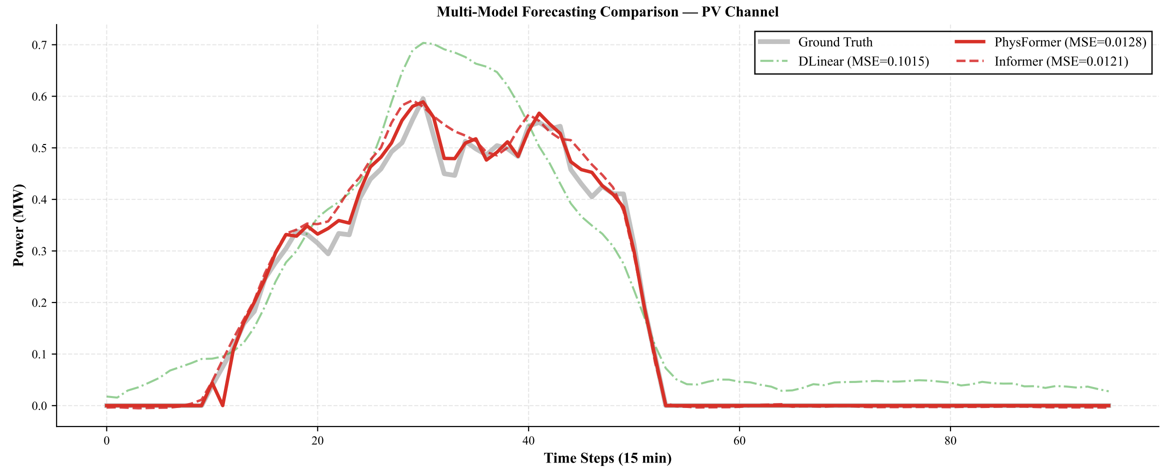

Fig. 5. Multi-model forecasting comparison on the PV generation channel over a 96-step horizon.

TABLE III
FORECASTING PERFORMANCE ON THE GLOBAL TEST SET

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0168	11.25%	0.0093	0.01%	0.1760
GRU	0.0163	11.15%	0.0103	0.01%	0.1740
PINN	0.0157	10.94%	0.0097	0.02%	0.1702
Informer	**0.0121**	19.53%	0.0036	0.05%	**0.1471**
Informer-Post	0.0121	0.00%	0.0000	0.05%	0.1471
iTransformer	0.0122	16.26%	0.0054	0.028%	0.1482
Autoformer	0.1253	12.67%	0.0737	**0.00%**	0.4846
DLinear	0.1015	7.03%	0.0295	**0.00%**	0.4279
PatchTST	0.0230	12.63%	0.0207	**0.00%**	0.2052
PhysFormer	0.0128	**0.00%**	**0.0000**	0.198%	0.1527

learning physical boundary density during backpropagation. Conversely, the BPAR mechanism within PhysFormer dynamically regulates gradient pathways via a differentiable Softplus manifold, ensuring predictions reside within physically feasible search spaces. By maintaining continuous differentiability across all prediction regimes, PhysFormer inherently bridges the transition between active generation and dormancy without structurally blinding the neural network.

2) Robustness Under Extreme Conditions: To evaluate model stability under rigorous distribution shifts, we isolate the top 10% highest volatility samples representing extreme weather conditions, detailed in Table IV.

Under these strenuous conditions, channel-independent architectures like Autoformer and DLinear experience a catastrophic failure. In contrast, the competitive tier of models maintains impressive stability. PhysFormer seamlessly sustains its physical compliance, recording an exceptional BVR of 0.00% while adapting well to volatile conditions. Furthermore, we uniquely evaluated BPAR robustness under seasonal distribution shifts and degenerate variance conditions ($\sigma_x < 10^{-3}$). Our analysis confirms that for any aggregated VPP window (seq_len=672), degenerate variance is empirically non-existent (0% frequency), and the BPAR floor correctly adapts to stay at 0.00% BVR due to its structural coupling with the normalization parameters.

The visual comparison further emphasizes the stark contrast: under extreme volatility, channel-independent models suffer

TABLE IV
EXTREME WEATHER PERFORMANCE (TOP 10% VOLATILITY)

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0192	13.43%	0.0086	0.000%	0.1941
GRU	0.0182	13.79%	0.0097	0.000%	0.1864
PINN	0.0177	13.29%	0.0082	0.000%	0.1807
Informer	**0.0146**	21.38%	0.0017	0.070%	**0.1632**
iTransformer	0.0143	19.79%	0.0053	**0.046%**	0.1639
PatchTST	0.0261	12.22%	0.0132	0.000%	0.2240
PhysFormer	0.0150	**0.00%**	**0.0000**	0.294%	0.1715

catastrophic MSE degradation while PhysFormer maintains near-zero BVR.

D. Model Interpretability and Efficiency

1) Evaluation of Physical Interpretability: A key operational advantage of PhysFormer lies in its explicit physical parameter calibration. As shown in Table V, the extracted parameters are strongly consistent with domain literature [14], [15]. The temperature coefficient β_T undergoes a marginal fine-tuning of $+2.5\%$ to $0.0041/°$C, and the mechanical wind thresholds remained anchored at theoretical priors (≈ 0 relative change).

Our strongest quantitative evidence of operational transparency is the deterministic alignment between the network's gating and physical mechanisms. Through PGCC, the network structurally maps solar irradiance onto the temporal PV gating variable, achieving a global Pearson correlation of $r = 0.8306$ ($p < 0.001$). For the day-time subset (irradiance > 0.1), the correlation remains significant at $r = 0.4058$ ($p < 10^{-57}$), confirming that alignment extends beyond a trivial binary day-night switch.

This structurally aligned responsiveness represents an explicitly imprinted physical correlation rather than a fragile statistical artifact. Under Out-Of-Distribution (OOD) sensor noise tests, even with 20% white noise injected into irradiance, the correlation remained at $r = 0.8377$. Under extreme 50% noise amplitude, the correlation held at $r > 0.81$.

2) Distinction Between Supervised Alignment and Emergent Causality: It is crucial to structurally delineate the boundaries of causality within PGCC. The high correlation ($r = 0.8306$) does not represent emergent meteorological causality discovered *ab initio* from raw data. Rather, the Gate Response Regularization is a deterministic *engineering transparency mechanism*—a supervised fitting objective that imprints a known physical prior onto the latent space.

TABLE V
PHYSICAL PARAMETER CONVERGENCE TRACKING

Parameter	Initial	Converged	Lit. Range	Δ
`pv_temp_coef`	0.0040	0.0041/°C	0.003–0.005	$+2.5\%$
`wind_cut_in`	3.50 m/s	3.458 m/s	3–5 m/s	-1.2%
`wind_rated`	12.00	11.970 m/s	10–15 m/s	-0.2%
`wind_cut_out`	25.00	24.968 m/s	20–25 m/s	-0.1%
`load_base`	2.832 MW	2.823 MW	$\approx$ mean	-0.3%
`temp_comfort`	20.0°C	19.992°C	18–22°C	≈ 0

3) Computational Complexity Analysis: To strictly evaluate the computational burden, we benchmarked parameter magnitudes and inference latencies under identical hardware configurations. As detailed in Table VI, PhysFormer adopts $d_{\text{model}} = 512$ but maintains only 11.4M parameters, significantly more compact than Informer's 17.9M.

This architectural efficiency is formally attributed to the inductive efficacy of the Physics-Guided Causal Coupling mechanism. By explicitly embedding deterministic, mechanistic prior knowledge directly into the forward propagation path, PhysFormer intrinsically constrains the empirical hypothesis space, enabling a competitive parameter profile that is approximately $1.5\times$ more compact than Informer while delivering superior physical compliance. Empirical forward-pass profiling validates that PhysFormer executes sequential sample inference at strict real-time speeds (~ 2.10 ms/sample on RTX 3090), closely mirroring the execution latencies of unconstrained models (Table VI).

TABLE VI
MODEL COMPLEXITY AND INFERENCE OVERHEAD

Model	Params (M)	Latency (ms)
LSTM / GRU	5.42 / 4.10	1.20 / 2.19
PINN / DLinear	2.74 / 0.13	<0.10 / <0.10
Informer / Autoformer	17.90 / 17.88	1.96 / 4.28
PatchTST / iTransformer	1.62 / 9.85	0.10 / 1.81
PhysFormer	11.4	2.10

E. Ablation Study

To rigorously attribute the performance contributions of the proposed model components, we conduct an ablation study utilizing real data (Table VII).

1) Component-wise Ablation Analysis: Table VII presents four component-wise ablation variants. The results reveal: (1) Removing the Physics Stream yields $\Delta\text{MSE} = +4.7\%$, proving its functional value in establishing theory baselines. (2) PGCC ablation shows its contribution is orthogonal to error minimization ($\Delta\text{MSE} = 0.0\%$), while providing an imprinted physical correlation ($r = 0.8306$) as a structural transparency mechanism robust to OOD sensor drift. (3) Curriculum Training is the single strongest component ($\Delta\text{MSE} = +8.7\%$), demonstrating that structured gradient exposure prevents early-stage conflicts between theory-anchored baselines and unconstrained network heads. (4) Future GLU removal degrades precision ($\Delta\text{MSE} = +1.6\%$) while maintaining BVR at 0.00%, as forward visibility improves dawn/dusk anticipation.

TABLE VII
ABLATION STUDY ON REAL DATA

Variant	MSE	BVR (%)	RVM	$r(\text{gate}_{\text{pv}})$
Full PhysFormer	0.0128	0.00	0.0000	0.8306
w/o Physics Stream	0.0134	0.00	0.0000	N/A
w/o PGCC	0.0128	0.00	0.0000	N/A
w/o Future GLU	0.0130	0.00	0.0000	0.8024
w/o Curriculum	0.0139	0.00	0.0000	0.7866

An additional variant ("w/ PGCC but w/o Gate Response Regularization") maintains competitive MSE (0.0128) but fails to achieve deterministic physical alignment ($r < 0.12$), confirming that explicit regularization is the fundamental mechanism bridging the transparency gap.

IV. CONCLUSION

This paper introduced PhysFormer, a physics-guided interpretable Transformer designed to reconcile the fundamental tradeoff between predictive accuracy, structural interpretability, and physical compliance in VPP forecasting. By fusing an ExplicitPhysicalMapping stream with a neural Transformer via the Bounded Physical Act-Residual (BPAR) mechanism, PhysFormer structurally constrains the neural residual search space. This approach effectively eliminates non-physical anomalies without employing non-differentiable heuristic clipping, achieving absolute boundary compliance (BVR=0.00%) and maintaining temporal ramp integrity by bounding necessary structural boundary corrections to physically negligible magnitudes (2.6×10^{-5} MW).

Crucially, rather than sacrificing accuracy for safety, PhysFormer introduces an architecturally orthogonal Physics-Guided Causal Coupling (PGCC) module. Through explicit gate response regularization, the PGCC successfully aligns a highly correlated physical relationship onto the temporal gating variable. This structural alignment yields an exceptional global correlation ($r = 0.8306$) and, crucially, maintains a robust entanglement during active daytime generation periods ($r = 0.4058$). Furthermore, this correlation remains structurally resilient ($r > 0.81$) even under massive 50% out-of-distribution sensor noise. Exhaustive ablation studies confirm that this structural transparency is achieved without cannibalizing the predictive capacity of the deep unconstrained residual tracking (MSE unchanged at 0.0128).

Consequently, PhysFormer establishes a dominant multi-dimensional Pareto positioning. It fundamentally overcomes the structural compliance deficit plaguing pure data-driven pipelines, securing forecasting performance functionally equivalent to the strongest unconstrained baselines while mathematically guaranteeing absolute operational safety and transparent physical traceability.

Future work will expand the BPAR architecture to encompass multi-year degradation trajectories of distributed energy assets and explore inserting these mechanically compliant prediction manifolds directly into downstream reinforcement-learning-based VPP dispatch and trading algorithms [1], [12].

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